Process & Decision Document

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Role(s): Coder & Promter of AI changes

Primary responsibility for this work: Worked independently and with the Visual Studio Chatbot to come up with modifications for the code that will add a debug screen to the code that includes a toggle for moon gravity and toggle for collision boxes.

Goal of Work Session: During this work session, I focused on working with AI to integrate a debugging screen with two features on it that can be toggled on and off.

GenAI tools (if used):

- Chatbot on Visual Studio Code 6

GenAI Documentation

Date Used: March 19, 2026

Tool Disclosure: Chatbot on Visual Studio Code 6

Purpose of Use: To help code parts that were too complex for my skills.

Summary of Interaction:

- Prompt 1: I explained to the chatbot that I needed a debug screen that has a toggle for moon gravity and explained what that meant
- Prompt 2: Asked the chatbot to fix the error that it had not included the debug screen
- Prompt 3: Instructed it to add another feature of collision boxes to the debug menu

Human Decision Point(s):

- The debug screen was not working and I had to initiate the AI to make that change
- I made my instructions text at the top of the screen independently

```
sketch.js > draw
function draw() {
            player.vel.x = 1.5;
            player.mirror.x = false;
        }
    }

    // --- KEEP IN VIEW ---
    player.pos.x = constrain(player.pos.x, FRAME_W / 2, VIEWW - FRAME_

    // persistent debug button (always visible)
    camera.off();
    push();
    noStroke();
    fill(0, 120);
    rect(VIEWW - 44, 8, 36, 18, 4);
    fill(255);
    textSize(10);
    textAlign(CENTER, CENTER);
    text(debugMode ? "DBG ON" : "DEBUG", VIEWW - 26, 17);
    pop();
    camera.on();

    // Instruction text
    textSize(10);
    fill(0, 120);
    rect(4, 8, 100, 18, 4);
    fill(255);
    textAlign(LEFT, CENTER);
    text("M: DEBUG MENU", 10, 17);

    // --- DEBUG OVERLAY ---
```

My addition of the instructions of debugging

Integrity & Verification Note:

All decisions were verified using:

- My prior knowledge of coding from CS105 and CS106
- Knowledge of coding that was reviewed in Class 9 lecture by David

Scope of GenAI Use:

Specifically, GenAI did *not*:

- Create the instructions on screen for the debugging menu

Limitations or Misfires:

- The chatbot did not include an accessible debugging menu in after the first iteration and I had to re-prompt it to be added in.

Summary of Process (Human + Tool)
Three iterations with GenAI:

- Asked for a debug screen that has a toggle for moon gravity (tool)
- Asked the chatbot to add the debug screen that was not previously integrated (tool)
- Added the instruction text in (human, me)
- Requested another feature of collision boxes to be added to the debug menu (tool)

Decision Points & Trade-offs

Attempting to add my own debugging menu was making the code more complicated and messy so I made the trade-off of coding myself and worked with the chatbot to implement the menu and features instead. Overall, I was happy with my changes and how the chatbot helped me edit and make changes to the code.

Verification & Judgement

I evaluated the appropriateness of my changes by:

- My prior knowledge of coding from CS105 and CS106
- Knowledge of coding that was reviewed in Class 9 lecture by David

Limitations, Dead Ends, or Open Questions

- All issues were resolved successfully by the chatbot
- Minor issue did occur with the debugging menu not appearing after the first itteration

Appendix - Visual Studio Code 6 Chatbot

#1 Could you add a debug screen to the code that includes a toggle for moon gravity? This will allow players to switch gravity between normal earth gravity and moon gravity while the game is running.

#1 Response

The MCP server `io.github.github/github\-mcp\-server` may have new tools and requires interaction to start.

Start it now?

I'll search the workspace for the main sketch and related files to update the game gravity toggle.

I'll open sketch.js to locate where gravity is set and where to add a debug UI.

I'll create a short plan to add the debug screen and gravity toggle.

Now I'll apply a focused patch to sketch.js adding debug state, gravity toggle logic, keyboard and mouse handlers, and the debug overlay rendering.

Edited

sketch.js+54-0

I'll mark the implementation steps done in the todo list.

Done — added an on-screen debug overlay and a moon-gravity toggle.

- **What I changed:** Added debug UI and gravity toggle logic in sketch.js.
- **How to use:** Press F to show/hide the debug overlay. Press G or click the on-screen button (top-left) to toggle moon gravity.
- **Gravity values:** Earth = 10, Moon ≈ 1.62 (applies immediately at toggle).
- **Next step (optional):** I can run a quick local test (open index.html or run a Live Server) and verify behavior — want me to do that?
-

GPT-5 mini • 1x

sketch.js > ...

```
52    VIEWH = 180;
53
54  // tile width & height
55  const TILE_W = 24,
56    TILE_H = 24;
57
58  // size of individual animation frames
59  const FRAME_W = 32,
60    FRAME_H = 32;
61
62  // Y-coordinate of player start (4 tiles above the bottom)
63  const MAP_START_Y = VIEWH - TILE_H * 4;
64
65  // gravity
66  const GRAVITY = 10;
67  // moon gravity (approx. 1.62 m/s^2 relative to Earth's ~9.81)
68  const MOON_GRAVITY = 1.62;
69        You, 7 minutes ago • Uncommitted changes
70  // debug & gravity toggle state
71  let debugMode = false;
72  let moonGravityEnabled = false;
73
74  function preload() {
75    // --- IMAGES ---
76    playerImg = loadImage("assets/foxSpriteSheet.png");
77    bgImg = loadImage("assets/combinedBackground.png");
78    groundImg = loadImage("assets/groundTile.png");
79    groundDeepImg = loadImage("assets/groundTileDeep.png");
80
```

Keep Undo

```
function startMusicIfNeeded() {
    startLoop();
  }
}

function setMoonGravity(enabled) {
  moonGravityEnabled = !!enabled;
  world.gravity.y = moonGravityEnabled ? MOON_GRAVITY : GRAVITY;
}

function keyPressed() {
  startMusicIfNeeded();
  // toggle debug overlay
  if (key === 'f' || key === 'F') {
    debugMode = !debugMode;
  }
  // toggle moon gravity
  if (key === 'g' || key === 'G') {
    setMoonGravity(!moonGravityEnabled);
  }
}

function mousePressed() {
  startMusicIfNeeded();
  // If debug overlay is visible, allow clicking the gravity toggle box
  if (debugMode) {
    // toggle button region at top-left: x:8-112, y:8-40
    if (mouseX >= 8 && mouseX <= 112 && mouseY >= 8 && mouseY <= 40) {
      setMoonGravity(!moonGravityEnabled);
    }
  }
  // persistent debug toggle button (top-right)
  if (mouseX >= VIEWW - 44 && mouseX <= VIEWW - 8 && mouseY >= 8 && mouseY <= 26) {
    debugMode = !debugMode;
  }
}
```

Keep Undo 1 of 5

sketch.js > keyPressed

```
function draw() {
  camera.off();
  push();
  noStroke();
  fill(0, 120);
  rect(VIEWW - 44, 8, 36, 18, 4);
  fill(255);
  textSize(10);
  textAlign(CENTER, CENTER);
  text(debugMode ? 'DBG ON' : 'DEBUG', VIEWW - 26, 17);
  pop();
  camera.on();

  // --- DEBUG OVERLAY ---
  if (debugMode) {
    camera.off();
    push();
    noStroke();
    fill(0, 180);
    rect(4, 4, 160, 56, 6);
    fill(255);
    textSize(10);
    textAlign(LEFT, TOP);
    text("DEBUG", 12, 8);
    text(`Gravity: ${world.gravity.y.toFixed(2)}`, 12, 22);

    // draw toggle button
    const bx = 8, by = 34, bw = 104, bh = 16;
    stroke(255);
    strokeWeight(1);
    fill(moonGravityEnabled ? '#88c0ff' : '#444');
    rect(bx, by, bw, bh, 4);
    noStroke();
    fill(0);
    textAlign(CENTER, CENTER);
    text(moonGravityEnabled ? 'Moon Gravity (ON)' : 'Moon Gravity (OFF)', bx + bw / 2, by + bh / 2);

    pop();
```

#2 The debug screen isn't showing up could you add that please?

#2 Response

```
function keyPressed() {
  startMusicIfNeeded();
  // toggle debug overlay
  if (key === 'f' || key === 'F') {
    debugMode = !debugMode;
  }
  // toggle moon gravity
  if (key === 'g' || key === 'G') {
    setMoonGravity(!moonGravityEnabled);
  }
}

function mousePressed() {
  startMusicIfNeeded();
  // If debug overlay is visible, allow clicking the gravity toggle box
  if (debugMode) {
    // toggle button region at top-left: x:8-112, y:8-40
    if (mouseX >= 8 && mouseX <= 112 && mouseY >= 8 && mouseY <= 40) {
      setMoonGravity(!moonGravityEnabled);
    }
  }
  // persistent debug toggle button (top-right)
  if (mouseX >= VIEWW - 44 && mouseX <= VIEWW - 8 && mouseY >= 8 && mouseY <= 26) {
    debugMode = !debugMode;
  }
}
```

```
function draw() {

  // --- MOVEMENT ---
  if (!attacking) {
    player.vel.x = 0;
    if (kb.pressing("left")) {
      player.vel.x = -1.5;
      player.mirror.x = true;
    } else if (kb.pressing("right")) {
      player.vel.x = 1.5;
      player.mirror.x = false;
    }
  }

  // --- KEEP IN VIEW ---
  player.pos.x = constrain(player.pos.x, FRAME_W / 2, VIEWW - FRAME_W / 2);

  // persistent debug button (always visible)
  camera.off();
  push();
  noStroke();
  fill(0, 120);
  rect(VIEWW - 44, 8, 36, 18, 4);
  fill(255);
  textSize(10);
  textAlign(CENTER, CENTER);
  text(debugMode ? 'DBG ON' : 'DEBUG', VIEWW - 26, 17);
  pop();
  camera.on();

  // --- DEBUG OVERLAY ---
  if (debugMode) {
    camera.off();
    push();
    noStroke();
    fill(0, 180);
```

#3 Can you create collision boxes please and add it it as an option in the debug menu please?

Read sketch.js

Edited

sketch.js+2-0

Edited

sketch.js+4-0

Edited

sketch.js+4-0

Edited

sketch.js+38-0

Done — added collision boxes as a debug option in sketch.js.

What I added:

- showCollisionBoxes boolean flag.
- 'C' key toggle for collision boxes.
- Clickable "Collisions (ON/OFF)" button in the debug menu (below the moon gravity button).
- Rendering of collision rectangles when enabled: player (red), sensor (green), ground tiles (magenta).

How to use:

- Press 'M' to show the debug menu.
- Press 'C' or click the "Collisions" button in the menu to toggle collision box visibility.
- Boxes appear as outlines around sprites in the game world.

The debug overlay is now taller to fit both buttons. Moon gravity toggle remains as before.

Grok Code Fast 1 • 1x

sketch.js > ...

```
// Y-coordinate of player start (4 tiles above the bottom)
const MAP_START_Y = VIEWH - TILE_H * 4;

// gravity
const GRAVITY = 10;
// moon gravity (approx. 1.62 m/s^2 relative to Earth's ~9.81)
const MOON_GRAVITY = 1.62;

// debug & gravity toggle state
let debugMode = false;
let moonGravityEnabled = false;
// collision boxes debug
let showCollisionBoxes = false;
```

```
function keyPressed() {
  if (key === "g" || key === "G") {
    setMoonGravity(!moonGravityEnabled);
  }
  // toggle collision boxes
  if (key === "c" || key === "C") {
    showCollisionBoxes = !showCollisionBoxes;
  }
}

function mousePressed() {
  startMusicIfNeeded();
  // If debug overlay is visible, allow clicking the gravity toggle box
  if (debugMode) {
    // toggle button region at top-left: x:8-112, y:8-40
    if (mouseX >= 8 && mouseX <= 112 && mouseY >= 8 && mouseY <= 40) {
      setMoonGravity(!moonGravityEnabled);
    }
    // collision boxes toggle button: x:8-112, y:58-74
    if (mouseX >= 8 && mouseX <= 112 && mouseY >= 58 && mouseY <= 74) {
      showCollisionBoxes = !showCollisionBoxes;
    }
  }
  // persistent debug toggle button (top-right)
```

```
function draw() {
  // draw collision boxes in world space when enabled
  if (showCollisionBoxes) {
    // ensure we're drawing in world coordinates
    push();
    camera.on();
    noFill();
    strokeWeight(1);

    // player box (red)
    stroke(255, 100, 100);
    rectMode(CENTER);
    rect(player.pos.x, player.pos.y, player.w, player.h);

    // sensor box (green)
    if (typeof sensor !== 'undefined') {
      stroke(100, 255, 100);
      rect(sensor.pos.x, sensor.pos.y, sensor.w, sensor.h);
    }

    // ground tiles (magenta)
    stroke(200, 100, 200);
    if (ground && ground.length) {
      for (let i = 0; i < ground.length; i++) {
        const s = ground[i];
        if (!s) continue;
        rect(s.pos.x, s.pos.y, s.w || TILE_W, s.h || TILE_H);
      }
    }
    if (groundDeep && groundDeep.length) {
      for (let i = 0; i < groundDeep.length; i++) {
        const s = groundDeep[i];
        if (!s) continue;
        rect(s.pos.x, s.pos.y, s.w || TILE_W, s.h || TILE_H);
      }
    }

    pop();
  }
```

```
function setMoonGravity(enabled) {
}

function keyPressed() {
  startMusicIfNeeded();
  // toggle debug overlay
  if (key === "m" || key === "M") {
    debugMode = !debugMode;
  }
  // toggle moon gravity
  if (key === "g" || key === "G") {
    setMoonGravity(!moonGravityEnabled);
  }
  // toggle collision boxes
  if (key === "c" || key === "C") {
    showCollisionBoxes = !showCollisionBoxes;
  }
}

function mousePressed() {
  startMusicIfNeeded();
  // If debug overlay is visible, allow clicking the gravity toggle box
  if (debugMode) {
    // toggle button region at top-left: x:8-112, y:8-40
    if (mouseX >= 8 && mouseX <= 112 && mouseY >= 8 && mouseY <= 40) {
      setMoonGravity(!moonGravityEnabled);
    }
    // collision boxes toggle button: x:8-112, y:58-74
    if (mouseX >= 8 && mouseX <= 112 && mouseY >= 58 && mouseY <= 74) {
      showCollisionBoxes = !showCollisionBoxes;
    }
  }
```